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PART I

From Study 1 to Study 2: Roadmap and Design

1

The roadmap and what we did with it

1.1 What Study 1 left unresolved

Study 1 (Part II) ended with a strong benchmark-construction conclusion and a deliberately weak model-performance conclusion. Five problems stood between them.

1. **One question, six tests.** "Does crop-level splitting inflate accuracy?" was tested once per architecture inside a twelve-test Benjamini–Hochberg family. Every point estimate was positive, every test was underpowered, and none survived.
2. **Availability is not use.** A 424-way probe recognised fields far above chance, but that shows the shortcut is *available*, not that classifiers route their predictions through it.
3. **Leakage confounded with diversity.** At equal crop counts a crop-level training set contains about 914 distinct fields and a field-level one about 773; P1 differs from P2 in training diversity as well as in leakage.
4. **One rescored benchmark.** Only SIPaKMeD was rescored for classification; CRIC, whose crop-level split puts a training parent behind essentially every test cell, was not.
5. **The host.** The original classification matrix had to be repeated (Chapter 15 is the record), and runs were slower than expected for reasons only diagnosed in Study 2 (Section 26).

1.2 The roadmap, item by item

The October review turned these into a phased roadmap. Table 1.1 records what was done with every item, including the ones deliberately not done.

Table 1.1: The scaling roadmap and the response to each item.

Roadmap item	What was done	Status
Phase 0: arXiv readiness		
Fresh-clone check	Public repository cloned into a clean directory; its 45 tests pass. The public `main` branch stops at 22 August and lacks the Cx22 commits and a LICENSE file.	done; push and licence are authors' decisions

Roadmap item	What was done	Status
Text extraction of PDFs	Ligatures broke copy-paste in the drafts; `glyphtounicode` and T1 encoding added to every preamble.	done
Draft claims	The ISBI draft's sentence implying P3 is strictly more conservative than P2 flagged as an overreach; the draft's conclusion now says replication is in progress.	done
Phase 1: fix the statistics		
Single omnibus test	One crossed mixed model, `acc ~ x + (1+x|architecture) + (1|seed) + (1|split)`, with the split random effect that the roadmap's formula omitted (twelve architectures share each test set).	done (H1)
Simulation-based power	Custom R simulation of the exact model, 1,000 simulations per design cell, from the Study 1 variance components (Chapter 3).	done
More architectures	Twelve: the six Study 1 models plus ConvNeXt-T, Swin-T, RegNetY-1.6GF, EfficientNetV2-S, MaxViT-T, and self-supervised DINOv2 ViT-S/14.	done
15–25 seeds	Twenty, chosen by a registered power rule.	done
Preregistration	`PREREGISTRATION_STUDY2.md`, registered as Amendment 6 before any Study 2 model was trained.	done
Keep predictions, logits, checkpoints	Every run keeps per-crop logits, float16 final weights, and probe-split features; this is what made Amendment 8's evaluation-only arms possible.	done
Phase 2: causal arms		
2C leakage vs diversity	Factorial over one common test set at matched crop and field counts.	done (H3)
2A fingerprint suppression	Class-conditional parent adversary behind gradient reversal, seven-level λ sweep.	done (H5)
2B background transplant	Counterfactual composites with a same-parent (sibling) control.	done (H4)
Phase 3: generalisation		
Other domains (fish imagery)	Gated on Phase 1 and 2 outcomes; not started.	deferred by design
Part 5: datasets		
CRIC second rescore	Rescored with six architectures and ten seeds.	done (H2)
Keep Cx22 un-rescored	Kept; the official Cx22 split is parent-disjoint.	done
Cx22 literature check	Four open-access studies examined (Section 1.4).	done
JUCYT ground truth	Not publicly available.	not possible
Part 6 and Part 8		
Single-column neutral draft	`paper/v2`, every number generated.	done
Avoid scope creep	Study 2 ran only registered arms; Amendment 8 was added after Study 2's results with an explicit justification and its own Holm family.	done

1.3 Phase 0 in detail

The repository was cloned from its public remote into an empty directory, its environment created from the lock file, and its test suite run: 45 tests passed. Three gaps were recorded in `reports/study2/phase0_arxiv_checklist.md`. The public branch stops at 22 August 2026 and does not

contain the Cx22 segmentation commits; there is no LICENSE file, so nobody may legally reuse the toolkit; and several drafts produced PDFs whose ligatures (fi, fl) did not survive copy-paste. The third was fixed in every preamble. The first two are decisions for the authors and remain open.

1.4 The Cx22 literature check

The Cx22 release ships an official parent-disjoint split (Multi-Train/Multi-Test). The question was whether downstream users keep it. Of four open-access studies that train on Cx22, three use the official split. One describes image-level five-fold cross-validation over all 1,320 generated regions, which pools regions from the same CCEDD source image across folds. Following the project's no-naming rule for third-party work, only the aggregate is reported (`reports/study2/cx22_literature_check.md`).

1.5 Post hoc reanalysis of Study 1

While validating the analysis code, the registered H1 model was fitted to the sixty Study 1 SIPaKMeD P1 and P2 runs. It gives 1.37 pp (SE 0.38, Satterthwaite $df = 9.1$, $p = 0.005$). The same model on the Cx22 semantic-segmentation runs gives 0.16 Dice points ($p = 0.43$). Both analyses were specified after the Study 1 data were seen; they motivated Study 2 and supplied its pilot variances, and they are not evidence for Study 2 (`results/study2/posthoc/README.md`).

2

The Study 2 preregistration

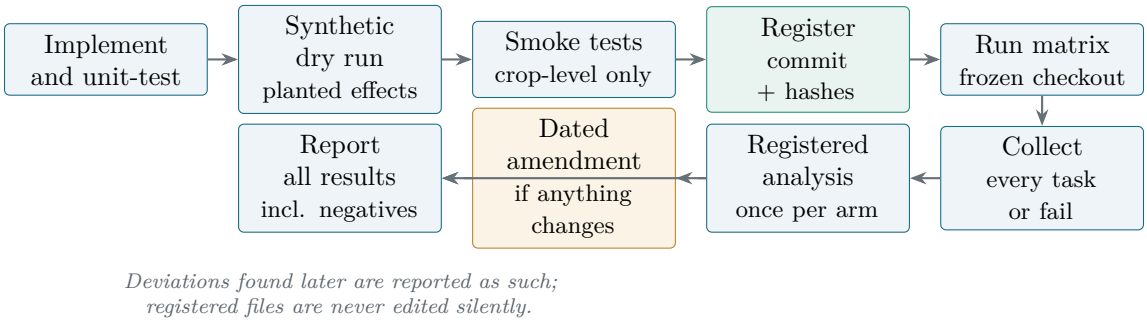


Figure 2.1: The governance loop used for Study 2 (Amendment 6), Amendment 7, Amendment 8, and Amendment 9. The registration commit fixes the analysis code, split manifests, task files, and cache hashes before any model whose result it governs is trained.

2.1 Status at registration

`PREREGISTRATION_STUDY2.md` was committed at `a62e8e7` (2026-10-18 05:07 IST) as Amendment 6 of the original plan. At that moment no Study 2 model had been trained for any registered cell. Before the commit: every split, design table, transplant plan, and input cache was generated and hashed; the full analysis pipeline was dry-run on a synthetic run matrix with planted effects (all recovered); and one-epoch smoke tests were run only on crop-level (P1) splits, so no protocol contrast could be observed (Table M.6).

2.2 What the plan fixes

Recipe
ImageNet initialisation, new linear head, AdamW (learning rate 10^{-4}, weight decay 10^{-4}), batch 64, 20 epochs, cross-entropy, final-epoch endpoint, no validation split, early stopping, or checkpoint selection; random resized crop (scale 0.8–1.0, ratio 3/4–4/3), horizontal and vertical flips, ImageNet normalisation.

Architectures
Twelve (Table 4.1).

Seeds
137, 173, 211, 257, 307, 349, 401, 457, 503, 557, 601, 653, 701, 751, 809, 853, 907, 953, 1009, 1051. Phase 1 uses all twenty, the mechanism arms and CRIC the first ten. The first five had been named as robustness seeds in the original plan and never run. No seed, architecture, or cell may be added or replaced after any result is seen.

Splits
Generated before registration under `splits/study2/` with SHA-256 hashes; trainers refuse any other.

H1 The crossed model on SIPaKMeD P1 and P2 accuracy; REML, bobyqa, Satterthwaite degrees of freedom, two-sided $\alpha = 0.05$; supported if $p < 0.05$ and positive. Singular fits are accepted; on convergence failure the model is simplified (uncorrelated slope, then no seed intercept) but the architecture slope and split effect are never dropped.

Assumption trigger
If no model converges, or residual |skewness| > 1, or excess kurtosis > 3, the primary result becomes a two-way cluster bootstrap (architectures and seeds resampled independently, 10,000 replicates).

Sensitivity
Kenward–Roger degrees of freedom, a binomial GLMM with an observation-level random effect, a seed-level exact sign-flip test, and the cluster bootstrap.

H2–H5
CRIC (H2), leakage at matched diversity (H3), the background context route (H4), and the gap's dependence on suppressed recoverability (H5), in one Holm family.

Exclusions
Runs are excluded only for logged infrastructure failure, non-finite loss, or a failed integrity check, never for magnitude or direction.

Gates
Gate A (H1) decides whether the inflation is reported as certified; Gate B decides whether the causal evidence is "clean" or "murky". Both decide framing only; every result is reported.

2.3 Amendment 7: failure handling

While Phase 1 ran, and before any 2A run had started, two cases the plan did not cover were registered (commit at 06:29 on 18 October). Infrastructure failures (another user's process exhausting GPU memory, loss of the host or controller, a filesystem error) do not count toward the one-retry limit. And because runs are deterministic, a gradient-reversal run whose loss becomes non-finite diverges again on retry; in that arm only, such a run is recorded as *diverged*, its λ leaves that architecture's regime, and it is excluded from H5. No Study 2 run failed or diverged, so neither provision was used.

3

Power analysis

3.1 Method

`analysis/study2/power_simulation.R` fits the H1 model to the sixty Study 1 runs and simulates it. The pilot gives a mean accuracy of 96.84%, an effect of 1.37 pp, architecture intercept and slope SDs of 0.44 and 0.36, split SDs of 0.31 (P1) and 0.59 (P2), a residual SD of 0.69, and a seed SD of zero. Six architectures put the intercept–slope correlation on its -1 boundary, so the data-generating process sets it to zero. Each design cell (seeds $\in \{10, 15, 20, 25, 30\}$, architectures $\in \{6, 8, 10, 12\}$, effect $\in \{0.5, 0.8, 1.0\}$ pp, variance scale $\in \{1, 1.5\}$) was simulated 1,000 times and fitted with the registered model.

3.2 The registered rule and the choice

The rule chose the smallest seed count, with twelve architectures, whose power for a 0.8-pp effect is at least 0.90 under pilot variances and at least 0.75 with every variance inflated by half. Twenty seeds give 0.977 and 0.782; fifteen fell short of the inflated criterion. A 0.8-pp effect exceeds the spread between published SIPaKMeD scores that motivates the question.

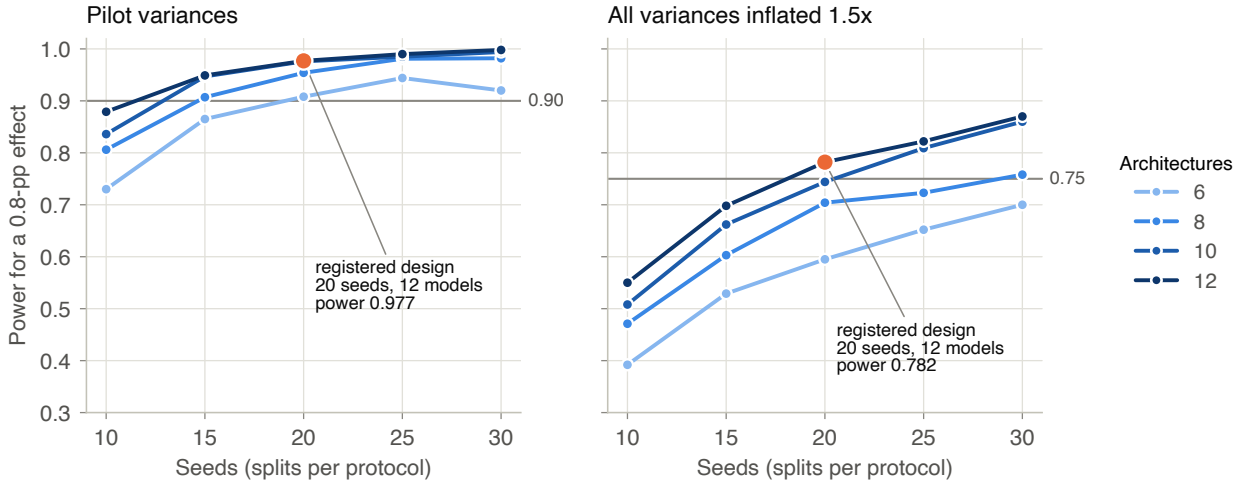


Figure 3.1: Simulated power of the H1 model for a 0.8-pp effect (1,000 simulations per design).

Table 3.1: Simulated power (1,000 simulations per cell) of the registered H1 model for a 0.8-pp effect, by variance scale, number of architectures, and number of seeds.

Variance	Archs	10 seeds	15 seeds	20 seeds	25 seeds	30 seeds
1×	6	0.730	0.865	0.908	0.944	0.920
1×	8	0.806	0.907	0.954	0.981	0.982
1×	10	0.836	0.947	0.976	0.985	0.994
1×	12	0.879	0.949	0.977	0.990	0.998
1.5×	6	0.392	0.529	0.595	0.652	0.700
1.5×	8	0.471	0.603	0.704	0.723	0.758
1.5×	10	0.508	0.662	0.744	0.809	0.860
1.5×	12	0.550	0.698	0.782	0.822	0.870

4

The designs in detail

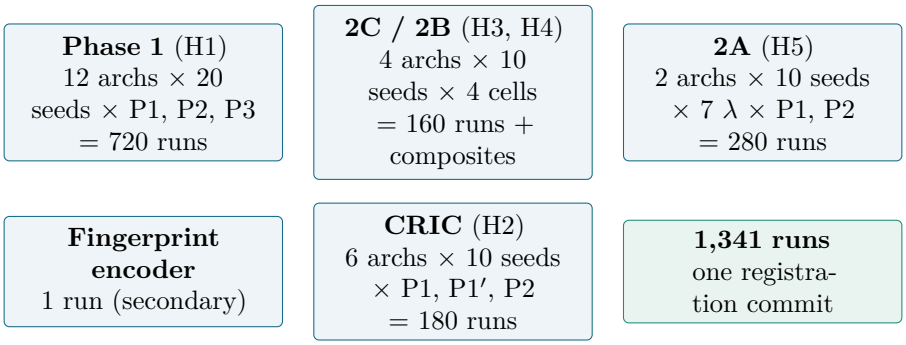


Figure 4.1: The registered Study 2 run matrix. P1-prime equals P1 byte for byte on every SIPaKMeD seed (P2 always has exactly 810 test crops), so it needed no separate SIPaKMeD runs.

4.1 Splits

P1 is a stratified 80:20 crop split; P2 a parent-disjoint split taken from `StratifiedGroupKFold` as the fold whose test size is closest to 810 crops; P3 the same on near-duplicate component groups. For every SIPaKMeD seed P2 has exactly 810 test crops, so P1-prime (a P1 split resampled to P2's size) is byte-identical to P1 and was not trained separately. For CRIC, P1-prime differs from P1 by 2–11 training cells and was trained.

4.2 Architectures

Table 4.1: The twelve Study 2 backbones: penultimate feature dimension, ImageNet weights, and mean Phase 1 training and total minutes per run.

Architecture	Dim	Weights	Train min	Total min
ResNet-18	512	ResNet18_Weights. IMAGENET1K_V1	0.85	0.87
ResNet-50	2048	ResNet50_Weights. IMAGENET1K_V2	2.60	2.64
DenseNet-121	1024	DenseNet121_Weights. IMAGENET1K_V1	3.25	3.29

Architecture	Dim	Weights	Train min	Total min
EfficientNet-B0	1280	EfficientNet_B0_Weights. IMAGENET1K_V1	1.78	1.80
MobileNet-V2	1280	MobileNet_V2_Weights. IMAGENET1K_V2	1.45	1.48
ViT-B/16	768	ViT_B_16_Weights. IMAGENET1K_V1	9.09	9.17
ConvNeXt-T	768	ConvNeXt_Tiny_Weights. IMAGENET1K_V1	5.15	5.21
Swin-T	768	Swin_T_Weights. IMAGENET1K_V1	4.61	4.66
RegNetY-1.6GF	888	RegNet_Y_1_6GF_Weights. IMAGENET1K_V2	2.47	2.50
EfficientNetV2-S	1280	EfficientNet_V2_S_Weights. IMAGENET1K_V1	3.46	3.50
MaxViT-T	512	MaxVit_T_Weights. IMAGENET1K_V1	7.76	7.86
DINOv2 ViT-S/14	384	timm:vit_small_patch14_ dinov2.lvd142m	4.08	4.13

4.3 2C: leakage at matched diversity

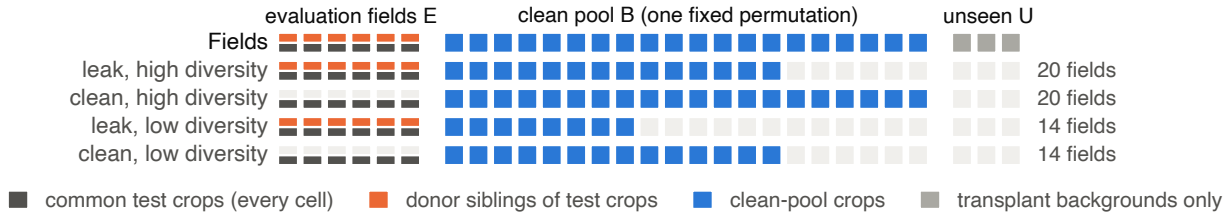


Figure 4.2: The 2C factorial for one seed: one common test set; leak cells add the evaluation fields' sibling crops, clean cells use other fields; within a diversity level crop and field counts are identical, and low-diversity field sets are nested in high-diversity ones.

For each seed and class, 30% of fields with at least two crops are evaluation fields; half of each one's crops (rounded down) form the common test set (about 527 crops) and the rest are its donor siblings. Six fields per class with at least three crops are reserved as unseen background donors for 2B. Each training set holds exactly 220 crops per class. The clean cells draw from non-evaluation fields along one fixed permutation; the leak cells replace the last $|E|$ fields of that permutation with the evaluation fields' donor siblings; the low-diversity cells use the first 70% of the permutation. High-diversity cells therefore contain 720 distinct fields and low-diversity cells 504. Every test crop has a training sibling in the leak cells (2.0 on average at high diversity, 3.0 at low) and none in the clean cells. A 240-crop budget was infeasible for seed 307 and for 10 of 100 unrelated seeds, so 220 was chosen before any training; it was feasible for 100 of 100.

4.4 2B: counterfactual background transplants

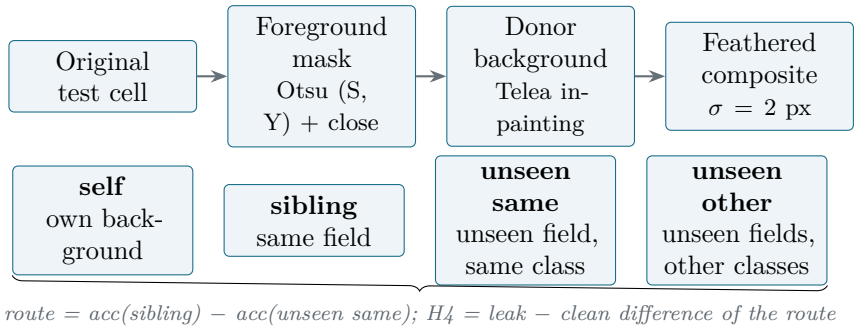


Figure 4.3: The 2B counterfactual background transplants. The *self* condition measures seam and inpainting artifacts; every contrast compares conditions that share them.

4.5 2A: adversarial fingerprint suppression

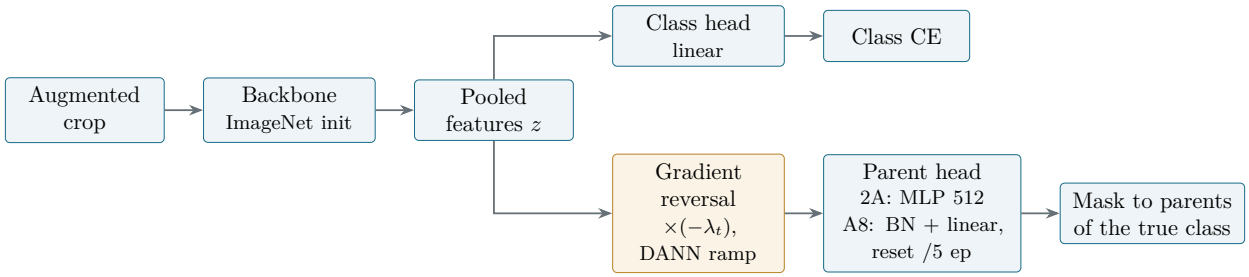


Figure 4.4: The adversarial suppression arms. The parent adversary only competes among parents of the sample's own class, so suppressing it never requires removing class information. Study 2 (2A) used a two-layer MLP; Amendment 8 (F7) used a probe-matched head (per-feature standardisation and one linear layer, its own optimiser at 10^{-3}, re-initialised every five epochs) so that features could not hide parent identity from it while leaving it readable by a linear probe.

The maximum reversal coefficient followed the DANN ramp $\lambda\,(2/(1+e^{-10p})-1)$ over training progress p, with $\lambda \in \{0, 0.01, 0.03, 0.1, 0.3, 1, 3\}$. Recoverability R is the within-class field-identification accuracy of a multinomial logistic regression (C = 1, standardised features) on frozen features of the balanced 424-field probe split (two fitting and two evaluation crops per field). The *regime* is the set of λ whose mean P2 accuracy stays within 1 pp of $\lambda = 0$; the *manipulation check* requires R at the strongest in-regime λ to fall by at least a quarter of its above-floor baseline.

4.6 CRIC

CRIC cells are 100×100 windows centred on the published nucleus coordinates (shifted inward at image borders) of the 400 Figshare slide images, whose MD5 and SHA-256 were verified (`data_manifests/cric_images_verified.json`). Six Bethesda classes; ResNet-18, ResNet-50, DenseNet-121, EfficientNet-B0, ConvNeXt-T, ViT-B/16; P1, P1-prime, P2; ten seeds.

5

Structure before training

For a parent with k crops allocated independently with test fraction q, the parent straddles the split with probability $1-(1-q)^k-q^k$; a test crop has a training sibling with probability $1-q^{k-1}$ and $(1-q)(k-1)$ training siblings on average; and the ratio of distinct training parents between a crop-level and a parent-level split of equal size is $\sum_j\{1-q^{k_j}\}/\{(1-q)m\}$ for m parents. Class–parent dependence is the mutual information between parent and class divided by the class entropy. `cytofair risk-report` computes all of them from any crop-to-parent table (`src/cytofair/risk.py`).

Table 5.1: `cytofair risk-report` closed forms for the four datasets at a 20% test fraction: crops per parent, expected straddle and sibling rates (percent), siblings per test crop, training-diversity ratio, and class–parent dependence (fraction of class entropy explained by the parent).

Dataset	Obs.	Parents	Obs./parent	Straddle	Test w/ sibling	Siblings	Diversity	Class dep.
SIPaKMeD	4,049	966	4.19	44.8	91.7	5.16	1.176	1.000
Cx22	1,320	514	2.57	33.9	83.6	1.95	1.164	–
CRIC	11,534	400	28.84	94.9	100.0	35.56	1.250	0.676
Herlev	917	917	1.00	0.0	0.0	0.00	1.000	1.000

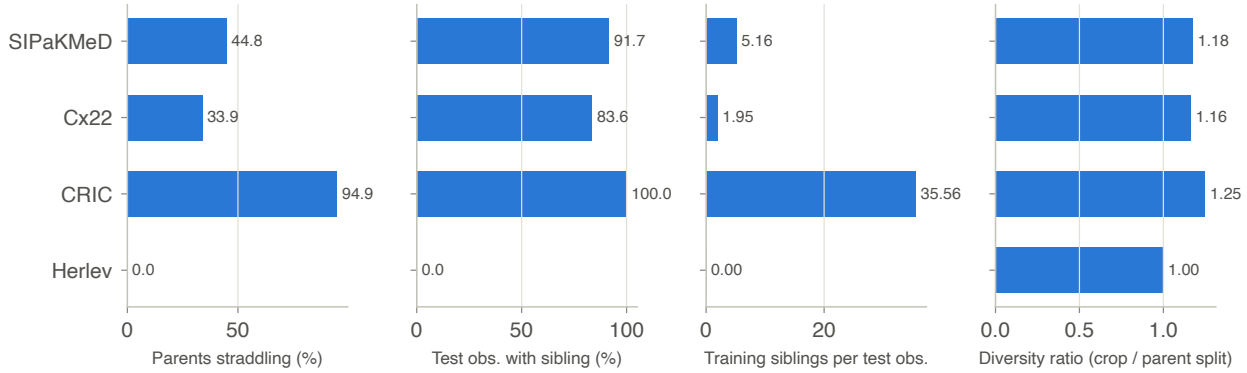


Figure 5.1: Closed-form structural exposure of the four datasets to a random 80:20 crop split.

Under five random P1 splits, the share of test observations sharing at least one source pixel with a training observation was 52.6% for SIPaKMeD, 71.5% for CRIC, and 34.0% for Cx22 (`results/study2/structural/literal_overlap_p1.csv`). CRIC is exposed on every measure: 29 cells per slide image, essentially every test cell with a training sibling, and slide identity explaining two thirds of the class entropy. That is the structure that predicted, before training, that CRIC's inflation would be the largest.

Bibliography